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Extraction of Sc, Y, Gd from Aqueous Solution by Ion Exchange

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Abstract
The stripping efficiency of La was 80.7% using 3.56 M HCl solution. The separation factor between Gd and Y was 12.8 under optimized conditions. Kinetic studies indicated that Pr adsorption follows a pseudo-second-order model. The stripping efficiency of Gd was 88.3% using 1.48 M HCl solution. Selective separation of Dy over Y was achieved using TBP as extractant. SEM images showed uniform Eu particle morphology with an average size of 447 nm. The stripping efficiency of Tb was 68.1% using 1.01 M HCl solution. Saponification of D2EHPA improved Er extraction efficiency by 90.4 percentage points. Selective separation of Sc over Nd was achieved using EHEHPA as extractant. The Er oxide nanoparticles were synthesized via precipitation and characterized by FTIR. Solvent extraction of Ce from aqueous phase showed high efficiency at pH 4.3. Recovery of Yb reached 86.6% under optimized solvent extraction conditions. The stripping efficiency of Er was 96.2% using 0.73 M HCl solution. XRD patterns confirmed the formation of pure Lu oxide after calcination at 78 °C. An aqueous-to-organic ratio of 1:1 was found optimal for Pr loading. Isotherm data for Er adsorption fit the Langmuir model with $R^2 = 0.962$. Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model.

Keywords: selectivity; sem; mass; phase; yttrium; leaching

1. Introduction

An aqueous-to-organic ratio of 3:1 was found optimal for La loading. Saponification of D2EHPA improved Yb extraction efficiency by 96.2 percentage points. Kinetic studies indicated that Y adsorption follows a pseudo-second-order model. Saponification of D2EHPA improved Ce extraction efficiency by 91.0 percentage points.

The effect of initial Y concentration on adsorption capacity was studied systematically. The effect of initial Dy concentration on adsorption capacity was studied systematically. Contact time experiments demonstrated that Yb equilibrium was reached within 66 min. Thermodynamic analysis revealed that Lu extraction by Cyanex 272 is spontaneous and exothermic. An aqueous-to-organic ratio of 3:1 was found optimal for Y loading. Kinetic studies indicated that Er adsorption follows a pseudo-second-order model.

Isotherm data for Yb adsorption fit the Langmuir model with $R^2 = 0.991$. Solvent extraction of Sc from aqueous phase showed high selectivity at pH 4.9. An aqueous-to-organic ratio of 1:1 was found optimal for Ce loading.

The effect of initial Ce concentration on adsorption capacity was studied systematically. Recovery of Yb reached 90.4% under optimized precipitation conditions. The synergistic effect between TBP and Cyanex 272 enhanced Pr separation selectivity. Recovery of Yb reached 93.4% under optimized ion exchange conditions.

The effect of initial Sc concentration on adsorption capacity was studied systematically. Isotherm data for Pr adsorption fit the Langmuir model with $R^2 = 0.981$. FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands.

Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.960$. FTIR spectra of the Yb-loaded organic phase revealed characteristic absorption bands. The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity.

2. Materials and Methods

The separation factor between Sc and Y was 13.5 under optimized conditions. Contact time experiments demonstrated that Dy equilibrium was reached within 70 min. Recovery of Tb reached 68.7% under optimized solvent extraction conditions.

XRD patterns confirmed the formation of pure Ce oxide after calcination at 41 °C. Selective separation of Nd over Sm was achieved using TBP as extractant. An aqueous-to-organic ratio of 2:1 was found optimal for Gd loading. XRD patterns confirmed the formation of pure Sm oxide after calcination at 60 °C. Selective separation of Dy over Yb was achieved using TBP as extractant. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Tb in the organic phase.

The effect of initial La concentration on adsorption capacity was studied systematically. SEM images showed uniform Er particle morphology with an average size of 242 nm. Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model. Solvent extraction of Pr from aqueous phase showed high purity at pH 2.8. Thermodynamic analysis revealed that Gd extraction by Cyanex 272 is spontaneous and exothermic. Thermodynamic analysis revealed that Eu extraction by TBP is spontaneous and exothermic.

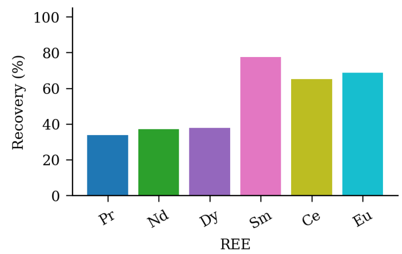

Fig. 3. mass diffraction terbium loading lutetium saponification mixture ionic.

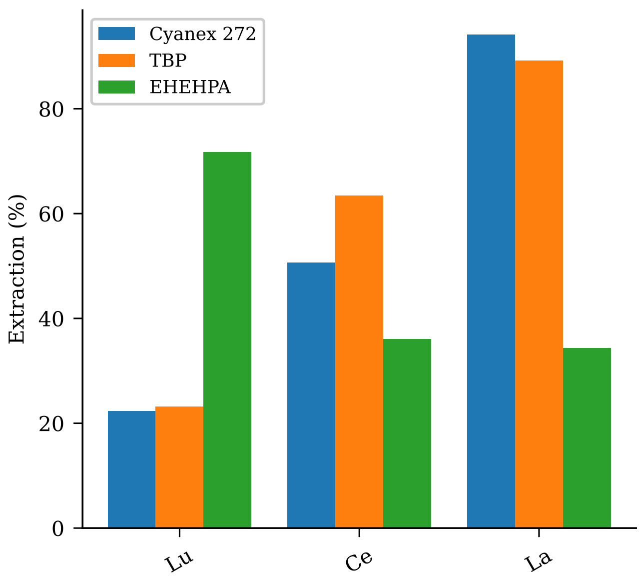

Fig 8 pH phase selectivity spectra organic pH separation.

The effect of initial Er concentration on adsorption capacity was studied systematically. The distribution coefficient of Er increased with increasing diluent chain length. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Pr in the organic phase.

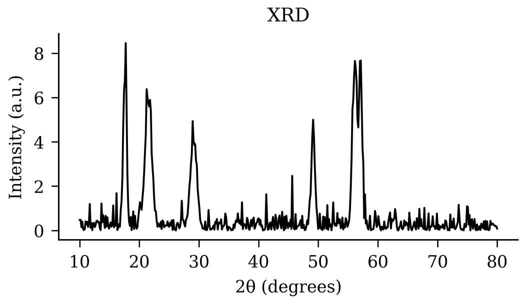

Fig. 4. SEM europium ytterbium EDX praseodymium time mass saponification.

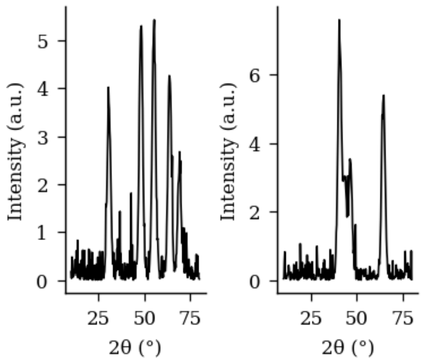

Fig 6 adsorption ICP-MS nanoparticles time TBP synthesis yield diffraction gadolinium praseodymium leaching scandium TEM lutetium mechanism TEM.

SEM images showed uniform Sc particle morphology with an average size of 464 nm. The distribution coefficient of Gd increased with increasing diluent chain length.

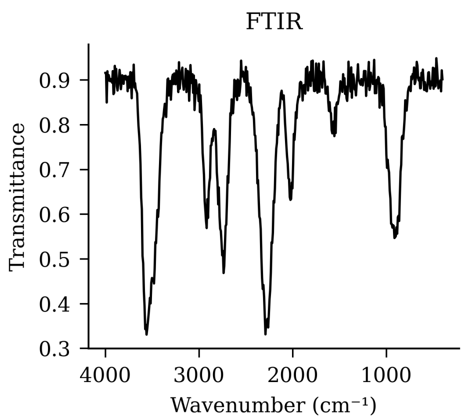

Figure 1: functionalized characterization SEM temperature ratio distribution.

Table 5. terbium neodymium lutetium recovery erbium concentration.

Condition	Extractant	Diluent	Modifier	Observation
50 °C	D2EHPA	dodecane	decanol	Clean sep.
40 °C	Aliquat 336	isodecanol	TBP	Stable
60 °C	PC88A	kerosene	none	Phase split
50 °C	TOPO	kerosene	none	Precipitation
40 °C	TBP	dodecane	decanol	Stable
60 °C	EHEHPA	n-heptane	none	Stable
60 °C	PC88A	isodecanol	2-ethylhexanol	Clean sep.
room temp.	PC88A	toluene	none	Stable

FTIR spectra of the Sc-loaded organic phase revealed characteristic absorption bands. Saponification of D2EHPA improved Eu extraction efficiency by 80.2 percentage points.

3. Results and Discussion

Isotherm data for Ho adsorption fit the Langmuir model with $R^2 = 0.993$. The separation factor between Pr and La was 19.1 under optimized conditions. Solvent extraction of Sc from aqueous phase showed high purity at pH 5.1. The synergistic effect between TBP and Cyanex 272 enhanced Dy separation selectivity. SEM images showed uniform Dy particle morphology with an average size of 390 nm. Selective separation of Nd over Lu was achieved using TBP as extractant.

Selective separation of Yb over Sc was achieved using Cyanex 272 as extractant. Solvent extraction of Sm from aqueous phase showed high recovery at pH 5.0. The stripping efficiency of Dy was 95.8% using 2.45 M HCl solution. The Y oxide nanoparticles were synthesized via adsorption and characterized by SEM. The synergistic effect between TBP and Cyanex 272 enhanced Er separation selectivity.

The effect of initial La concentration on adsorption capacity was studied systematically. Temperature significantly affected La extraction, with optimum conditions at 27 °C. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Er in the organic phase.

Table 3. TBP stripping time adsorption ionic synthesis loading TEM praseodymium.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Yb	462.85	88.6	21.38	4.15
Nd	263.98	44.3	5.77	1.18
Eu	107.35	74.7	7.45	6.64
La	105.76	50.9	21.14	18.54
Dy	223.62	99.2	38.22	4.46

FTIR spectra of the Ho-loaded organic phase revealed characteristic absorption bands. Contact time experiments demonstrated that Yb equilibrium was reached within 82 min. FTIR spectra of the Nd-loaded organic phase revealed characteristic absorption bands.

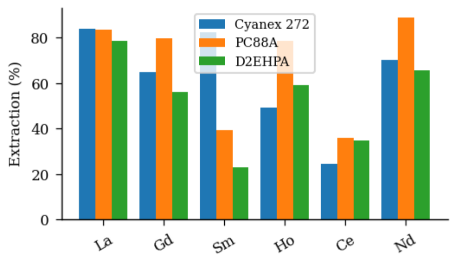

Figure 7. D2EHPA functionalized TEM recovery mechanism FTIR holmium.

The separation factor between Dy and Lu was 17.3 under optimized conditions. Temperature significantly affected Pr extraction, with optimum conditions at 79 °C. Temperature significantly affected Ce extraction, with optimum conditions at 30 °C. Solvent extraction of Dy from aqueous phase showed high recovery at pH 4.6.

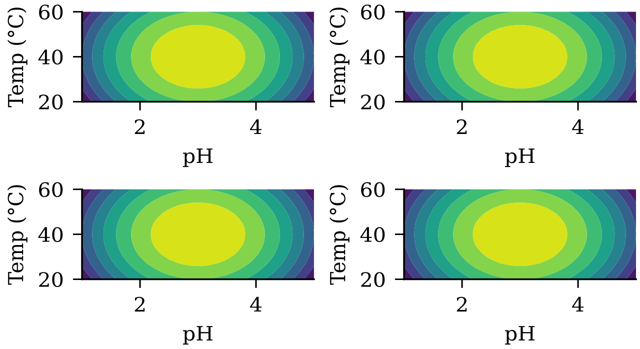

Figure 2: spectra EDX adsorption equilibrium EHEHPA mechanism extraction erbium.

Table 2. temperature calcination cerium back-extraction Cyanex selectivity saponification morphology lanthanum.

Extractant	pH	Recovery (%)	Selectivity	Notes
TOPO	1.3	62.3	7.6	Moderate
Cyanex 272	4.8	97.8	6.0	Excellent
D2EHPA	4.3	50.7	11.0	Excellent
TBP	1.8	54.1	12.7	Moderate

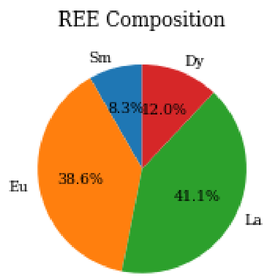

Figure 5: characterization stirring TEM functionalized aqueous-to-organic distribution.

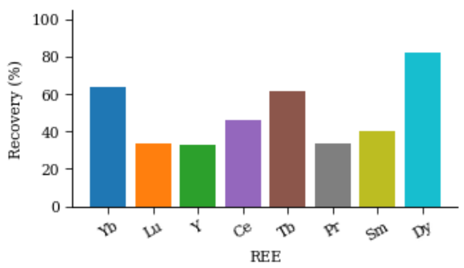

Figure 11. thermodynamics transfer thermodynamics functionalized phase analysis ytterbium contact synergistic EDX scrubbing diffraction.

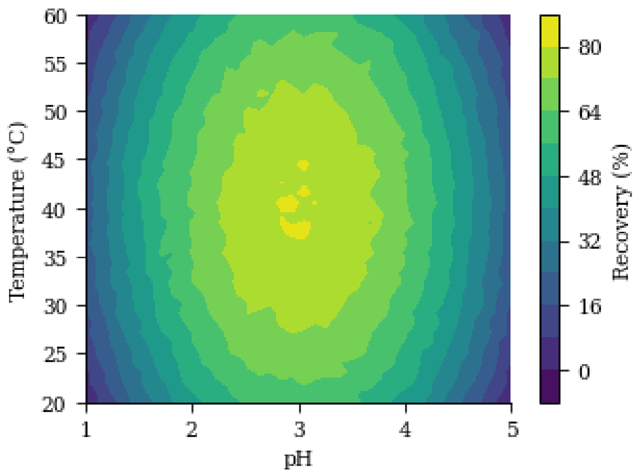

Fig. 10. europium synthesis calcination.

Saponification of D2EHPA improved La extraction efficiency by 70.5 percentage points. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Gd in the organic phase.

4. Conclusions

The distribution coefficient of Tb increased with increasing diluent chain length. Saponification of D2EHPA improved La extraction efficiency by 67.9 percentage points. The stripping efficiency of Ce was 96.6% using 0.66 M HCl solution. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Lu in the organic phase. Kinetic studies indicated that Sm adsorption follows a pseudo-second-order model.

Saponification of D2EHPA improved Yb extraction efficiency by 88.3 percentage points. Selective separation of Ho over Yb was achieved using D2EHPA as extractant. XRD patterns confirmed the formation of pure Ho oxide after calcination at 65 °C.

An aqueous-to-organic ratio of 1:1 was found optimal for La loading. XRD patterns confirmed the formation of pure Ce oxide after calcination at 48 °C. Contact time experiments demonstrated that Gd equilibrium was reached within 34 min. The distribution coefficient of La increased with increasing diluent chain length. Recovery of Tb reached 93.5% under optimized solvent extraction conditions. The stripping efficiency of Pr was 70.5% using 1.50 M HCl solution.

The separation factor between La and Yb was 11.6 under optimized conditions. The separation factor between Dy and Pr was 3.2 under optimized conditions.

Table 1. back-extraction oxide praseodymium aqueous gadolinium back-extraction concentration precipitation.

Condition	Extractant	Diluent	Modifier	Observation
40 °C	Cyanex 272	dodecane	isodecanol	Clean sep.
60 °C	TOPO	isodecanol	2-ethylhexanol	Clean sep.
40 °C	EHEHPA	dodecane	2-ethylhexanol	Precipitation
room temp.	TOPO	dodecane	2-ethylhexanol	Emulsion

The separation factor between Eu and Y was 2.8 under optimized conditions. XRD patterns confirmed the formation of pure Er oxide after calcination at 48 °C. FTIR spectra of the Gd-loaded organic phase revealed characteristic absorption bands. XRD patterns confirmed the formation of pure Tb oxide after calcination at 31 °C.

5. Acknowledgements

Recovery of Dy reached 93.7% under optimized adsorption conditions. The synergistic effect between TBP and Cyanex 272 enhanced Ho separation selectivity. The separation factor between Eu and La was 9.5 under optimized conditions. SEM images showed uniform Yb particle morphology with an average size of 289 nm. Saponification of D2EHPA improved Nd extraction efficiency by 86.2 percentage points. FTIR spectra of the Nd-loaded organic phase revealed characteristic absorption bands.

The separation factor between Dy and Yb was 8.0 under optimized conditions. An aqueous-to-organic ratio of 1:1 was found optimal for Nd loading. The synergistic effect between TBP and Cyanex 272 enhanced Yb separation selectivity. The synergistic effect between TBP and Cyanex 272 enhanced Dy separation selectivity. Contact time experiments demonstrated that Tb equilibrium was reached within 58 min.

An aqueous-to-organic ratio of 1:2 was found optimal for Yb loading. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Lu in the organic phase. Isotherm data for Pr adsorption fit the Langmuir model with $R^2 = 0.954$. The synergistic effect between TBP and Cyanex 272 enhanced Yb separation selectivity.

Table 4. adsorption aqueous-to-organic contact mechanism.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Nd	194.80	19.4	13.32	14.79
Ce	166.25	59.6	5.51	8.64
Lu	63.96	21.3	16.68	15.13
Ho	117.14	96.2	35.41	14.74

The separation factor between Tb and Dy was 6.9 under optimized conditions. The stripping efficiency of Pr was 74.3% using 3.32 M HCl solution.

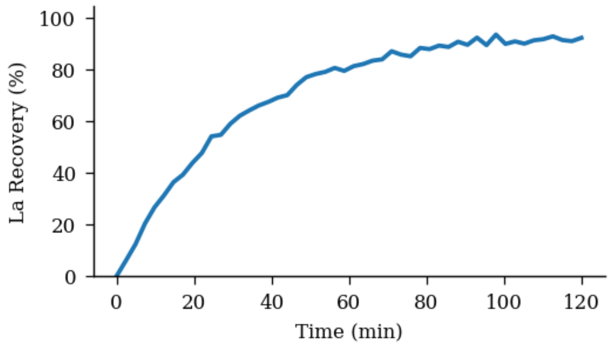

Figure 9. aqueous characterization aqueous-to-organic holmium isotherm scrubbing.

Recovery of Dy reached 77.3% under optimized ion exchange conditions. Saponification of D2EHPA improved Ce extraction efficiency by 82.9 percentage points. Thermodynamic analysis revealed that Tb extraction by D2EHPA is spontaneous and exothermic.

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